Music Practice Robot · Hardware Technical Summary

> Product Form: Desktop music practice robot (screen + single-servo head + dual-microphone array) Core Features: Voice chat, sound-source localization, real-time performance-graph comparison, cloud upload of performance recordings, cloud GPT performance-review playback

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1. System Overview

The robot is built on the **ESP32-P4** high-performance MCU platform, forming a five-layer hardware architecture: **Compute — Sense — Display — Actuate — Power**.

- **Compute Layer**: ESP32-P4 (dual-core RISC-V) + wireless co-processor, responsible for audio processing, graph rendering, AI interaction, and cloud communication
- **Sensing Layer**: Dual-microphone array + ES7210 audio front end (capture, echo cancellation, sound-source localization)
- **Display Layer**: 4.3-inch 480×800 capacitive touch IPS screen (standard graph display, real-time performance track drawing, chat interface)
- **Actuation Layer**: 20kg high-torque 5V servo (head rotation, facing the sound source / interactive gestures)
- **Power Layer**: 3.7V lithium battery (MX1.25 connector), Micro SD local storage expansion

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2. Core Compute Platform

2.1 Main Controller: ESP32-P4 (RISC-V 32-bit Dual-Core High-Performance MCU)

Item	Description
Core	Dual-core RISC-V 32-bit (HP cores, up to 400MHz) + low-power LP core
Compute positioning	Handles real-time audio DSP, dual-core parallelism (one core for the audio pipeline, one for UI/networking), and lightweight AI inference (ESP-DL)
AI extension	Supports AI instruction extensions; can run wake-word, noise reduction, and simple keyword recognition locally

Why this choice matters: - The dual-core architecture allows the **audio capture / pitch-detection pipeline** and the **screen rendering / network communication** to run on separate cores, preventing UI jank from affecting sampling

real-time performance — the key to the practice-comparison feature. - Compared with the ESP32-S3, the P4's MIPI-DSI/CSI interfaces and higher clock frequency leave headroom for screen refresh rate and camera expansion.

2.2 Wireless Connectivity: Wi-Fi 6 + BLE 5

- **Note**: The ESP32-P4 itself has no RF. Wi-Fi 6 / BLE 5 are provided by an onboard/co-packaged wireless co-processor (e.g., ESP32-C6).
- **Usage**:
- Wi-Fi 6: uploading performance recordings to the cloud, online updates of graphs/song library, streaming uplink for voice chat, downlink of GPT review feedback (low latency, congestion-resistant)
- BLE 5: future provisioning, remote control, and peripheral expansion (Bluetooth pedal / metronome)

2.3 Memory Configuration

Storage	Capacity	Purpose
PSRAM (in-package)	32MB	Frame buffers (double buffering for the 480×800 screen), audio ring buffers, recording cache
NOR Flash (in-package)	32MB	Firmware, built-in music graph library, prompts, offline word lists
Micro SD card	As needed	Local storage of long performance recordings, sheet-music resource expansion

The 32MB PSRAM is a highlight of this design: a single 480×800@16-bit frame is about 750KB, so double buffering + UI layers + recording buffers fit comfortably in PSRAM with no external SDRAM required.

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3. Display and Human-Machine Interaction

Item	Specification
Screen	4.3-inch IPS, 480×800 (portrait, suited to scrolling graphs)
Touch	Capacitive multi-touch
Interface	MIPI-DSI (high refresh, low CPU load, driven directly by the P4 display controller)

Feature mapping: - **Graph mode**: a vertically scrolling note waterfall; the standard graph and the real-time performance track are drawn as two channels on the same screen, with deviations color-coded (hit / sharp / flat) - **Chat mode**: voice waveform + text dialogue interface - The portrait 480×800 aspect ratio naturally suits sheet-music presentation where "the timeline flows downward"

4. Audio Subsystem

4.1 Capture Chain: Dual-Microphone Array + ES7210

Component	Role
Dual MEMS microphones	A two-element array with spaced placement, capturing two synchronized audio channels
ES7210	Multi-channel audio ADC + audio front-end processing: **echo cancellation (AEC), noise suppression, automatic gain control (AGC)**

Three key roles: 1. **Full-duplex voice chat**: while the robot's speaker is playing sound, the microphones keep listening; the ES7210's AEC removes the echo of its own playback, enabling a barge-in conversation experience — talking and listening at the same time, interruptible at any moment 2. **Sound-source localization**: using inter-channel time/phase difference (TDOA) with the ES7210's synchronized sampling channels, the azimuth of the sound source is estimated (roughly ±90° left/right); the result drives the servo to turn the head toward the performer 3. **Performance capture**: provides a clean mono mixed signal for pitch detection, reducing the impact of environmental noise on note recognition

4.2 Playback Chain: ES8311 + Speaker

Component	Role
ES8311	Low-power audio codec (DAC); outputs voice replies, demonstration audio, and prompts
Speaker	Audio reproduction

Division of labor: the ES7210 handles "listening" (ADC / AEC / noise reduction) while the ES8311 handles "speaking" (DAC playback). Together they form a complete full-duplex voice front end — Espressif's officially recommended pairing for the P4 voice solution.

4.3 Performance-Comparison Data Flow (Critical Chain)

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Dual microphones → ES7210 (AEC / noise reduction / ADC) → I2S → ESP32-P4
                                                              ├── Pitch detection (YIN/FFT) → note sequence → graph track ov
                                                              ├── Raw audio → PSRAM/SD recording buffer
                                                              └── TDOA sound-source azimuth → servo angle command
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5. Motion Actuation: 20kg High-Torque 5V Servo

- **Purpose**: single-degree-of-freedom rotation of the head (the screen + enclosure assembly, or the head module):
- **Turning the head toward the sound source** (the performer's position) after sound-source localization
- Anthropomorphic gestures during chat (nodding, shaking the head)
- **Drive method**: angle controlled by a PWM signal generated by the P4's LEDC/MCPWM peripherals
- **20kg·cm torque assessment**: ample margin for a desktop robot head load (screen + enclosure); crisp start/stop and reliable motion
- **Power supply note**: the servo's stall current can exceed 1A. It **must be powered separately from the controller's 3.3V/1.8V digital rails**; a dedicated 5V branch plus a bulk filter capacitor is recommended to prevent the servo's startup transient from drooping the bus voltage and browning out the ESP32-P4

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6. Power System

Item	Description
Battery	3.7V lithium battery (single-cell Li-Po/Li-ion), MX1.25 connector
Power tree (inferred)	Battery → power management (charging + protection) → 3.3V (controller/peripherals), 5V boost (servo/speaker amplifier)

Power budget highlights: - Main consumers: screen backlight (several hundred mA), Wi-Fi 6 transmission, servo peak current, speaker amplifier - Recommendations: specify battery capacity (mAh) to estimate runtime; confirm in the schematic whether the charging circuit supports charge-while-use

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7. Feature × Hardware Mapping Table

Product Feature	Dependent Hardware
Voice chat (barge-in capable)	Dual microphones + ES7210 (AEC) + ES8311 + speaker + Wi-Fi 6 (cloud LLM/ASR/TTS)
Sound-source localization	Dual-microphone array + ES7210 synchronized capture + ESP32-P4 TDOA computation
Head turning / anthropomorphic gestures	20kg servo (PWM driven)
Standard graph display	4.3" IPS touch screen + 32MB PSRAM frame buffer
Real-time performance track drawing	Dual microphones → ES7210 → P4 pitch detection → on-screen overlay rendering (dual-core separation guarantees real-time performance)
Performance recording upload	Microphone capture → PSRAM/SD buffering → Wi-Fi 6 upload
Cloud GPT performance-review playback	Recording upload → cloud GPT model analysis (pitch accuracy / rhythm / completeness) → feedback text downlink → ES8311 + speaker TTS playback
Song library / graph storage	32MB NOR Flash + Micro SD card